AI-Powered Accessible Learning and Assessment Platform for Visually Impaired O/L History Students in Sri Lanka

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***Abstract*— Visually impaired students participating in Sri Lankan General Certificate of Education (Ordinary Level) History examinations face significant accessibility barriers, as Braille examination scripts require manual transcription prior to evaluation while independent study is limited by the lack of curriculum-specific interactive learning tools. This paper presents an integrated AI-powered platform that addresses both challenges through two complementary components. The first component implements an end-to-end Braille-to-text conversion and automated evaluation pipeline, where scanned or photographed Braille answer scripts are processed using computer vision-based optical character recognition (OCR), including dot detection, spatial clustering, and Unicode Braille mapping, enabling accurate reconstruction of machine-readable text without the need for specialized hardware. The transcribed responses are then evaluated against reference answers generated by a fine-tuned Llama 3.1 8B Instruct model using a weighted multi-metric scoring framework that combines semantic similarity (Sentence-BERT), keyword overlap (TF-IDF), and token-level Jaccard similarity, along with a length normalization factor. The second component introduces an AI-powered, voice-oriented adaptive quiz engine designed for self-study, where curriculum-aligned multiple-choice and free-text questions are generated using a local large language model, learner responses are evaluated through Sentence-BERT-based semantic analysis, and adaptive progression is governed by a simplified item response theory model that dynamically adjusts question difficulty based on learner performance. Learner sessions and performance histories are stored in MongoDB, and the system is deployed as a modular FastAPI service designed for integration with speech-to-text and text-to-speech interfaces. Together, these components form a unified platform for both assessment and learning support, enhancing evaluation efficiency, consistency, and accessibility for visually impaired O/L History students.**

Keywords*— *Accessible assessment, adaptive learning, automated grading, Braille OCR, large language models, natural language processing, semantic similarity, Sentence-BERT, TF-IDF, visually impaired students.

I. INTRODUCTION

Inclusive education frameworks increasingly affirm that accessibility must extend beyond physical participation in schooling to encompass all stages of learning, assessment, and feedback. For visually impaired students studying O-Level History in Sri Lanka, this principle carries particular significance. Such students rely on Braille for written examination responses; however, the subsequent processes of reading, transcribing, and evaluating these responses typically require sighted intermediaries, resulting in delays, inconsistencies, and limited scalability within already resource-constrained educational institutions.

Independent study presents an equally significant challenge. Curriculum-specific assistive learning systems remain limited in the Sri Lankan context. Existing tools primarily support generic information retrieval, mathematical tutoring, or narrow examination assistance, but do not provide subject-aligned intelligent learning experiences for humanities education. O-Level History, which requires factual recall, event sequencing, interpretation of historical developments, and chapter-based practice, is therefore particularly underserved by current technologies.

This paper presents an integrated AI-powered platform designed to address both challenges. The platform comprises two tightly coupled backend components. The first component is a Braille-to-text conversion and automated answer evaluation pipeline that enables timely and consistent assessment of handwritten Braille examination responses without requiring specialized scanning hardware. Computer vision techniques are used to detect Braille dot patterns from scanned or photographed answer scripts, while spatial clustering groups these dots into cells, and Unicode Braille decoding produces plain-text transcriptions. A domain-fine-tuned large language model then generates reference answers,

and a hybrid scoring framework evaluates student responses across semantic, keyword, and lexical dimensions.

The second component is a voice-oriented adaptive quiz engine that functions as the educational reasoning layer behind a speech-first interface. It generates curriculum-aligned questions, evaluates descriptive answers using sentence-level semantic embeddings, manages adaptive difficulty progression through a simplified item response theory model, and persists learner state for longitudinal progress tracking. The system is designed to enable visually impaired learners to interact entirely through spoken input and audio feedback, while the backend transparently handles all educational logic.

The contributions of this work are fivefold. First, the proposed Braille OCR module demonstrates that accurate dot-pattern recognition from examination scripts can be achieved using standard computer vision methods without specialized hardware. Second, a fine-tuned Llama 3.1 8B model is shown to generate contextually and pedagogically relevant reference answers for Sri Lankan O-Level History questions. Third, a hybrid multi-metric evaluation framework is introduced, integrating semantic similarity, keyword-based matching, and lexical overlap into a unified and interpretable scoring mechanism. Fourth, a practical backend architecture is presented that combines language model-based question generation, semantic answer evaluation, and rule-based adaptive progression within a single FastAPI service. Fifth, the system is explicitly grounded in the accessibility and curricular needs of visually impaired learners, rather than treating assistive learning as a generic search or retrieval problem.

II. LITERATURE REVIEW

A. Accessibility and Inclusive Education in Sri Lanka

Karunanayake and Wijesinghe reported that visually impaired learners in Sri Lanka continue to face structural barriers in their educational pathways due to unequal access to resources, specialist support, and adapted learning materials [1]. Their work establishes the local relevance of accessible educational technologies and underscores the need for context-aware digital interventions. Senavirathna et al. addressed a complementary problem through the development of a low-cost talking calculator, demonstrating that speech-enabled interfaces can make otherwise inaccessible tools usable for visually impaired users. However, their work also highlights that such task-specific interventions do not adequately support content-rich academic tutoring [2].

B. Braille Recognition and Computer Vision

Early Braille recognition systems relied on tactile sensors or specialized scanners to capture the raised-dot structure of embossed paper. Antonacopoulos and Bridson proposed a vision-based approach using connected-component analysis to detect individual dots and geometric reasoning to group them into cells, demonstrating the feasibility of hardware-independent Braille recognition [3]. Subhi and Ali developed a recognition system based on morphological operations and template matching, achieving high accuracy on single-grade Braille inputs [4]. Nandini et al. further demonstrated that preprocessing quality significantly influences downstream character recognition accuracy, with dot detection reliability, row segmentation, and cell pairing identified as the primary sources of error in image-based pipelines [5].

C. Automated Short-Answer Evaluation

Mohler and Mihalcea benchmarked multiple similarity metrics for automatic short-answer grading and found that combining surface-level and semantic features consistently outperforms any single metric [6]. Sultan et al. proposed an alignment-based semantic similarity approach, achieving competitive results on benchmark datasets [7]. Sung et al. applied transformer-based models to the grading task, demonstrating that contextualized representations capture meaning more effectively than traditional bag-of-words approaches [8]. Sentence-BERT, introduced by Reimers and Gurevych, provides computationally efficient sentence-level embeddings suitable for similarity tasks without requiring pairwise fine-tuning, and its cosine similarity scores have become a standard baseline in educational natural language processing applications [9].

D. Large Language Models and Parameter-Efficient Fine-Tuning

Fine-tuning large language models for domain-specific tasks has become feasible through parameter-efficient methods. Hu et al. introduced LoRA, which adapts pre-trained models by injecting low-rank matrices into attention layers, enabling task-specific fine-tuning with a small fraction of trainable parameters [10]. Touvron et al. described the Llama family of open large language models, which are widely adopted as base models for domain-specific adaptation [11]. These approaches are particularly relevant for educational applications, where domain-specific terminology and factual precision are important, but full model retraining is computationally expensive and often impractical.

E. Voice-Based and Adaptive Educational Systems

Najaah and Thowfeek proposed a voice-based online examination system for visually impaired users, demonstrating the feasibility of spoken question delivery and spoken-answer capture in assessment settings [12]. Ayo discussed intelligent voice-based e-education as a broader framework for accessible learning environments [13]. Mackowski et al. presented accessible tutoring systems for mathematics, including an audio-tactile platform for visually impaired learners, demonstrating that adaptive educational support can be effectively designed for this population, although primarily within STEM domains [14], [15]. Yang and Taele described a personalized audio-based learning approach for blind students, emphasizing adaptive content delivery, though not specifically aligned with the Sri Lankan curriculum [16]. Tsouktakou et al. reviewed the broader application of artificial intelligence in the education of visually impaired learners, highlighting personalization and educational independence as key objectives [17].

A consistent gap across the literature is that voice interaction, adaptive tutoring, and AI-supported education for visually impaired learners have been studied largely in isolation, and rarely in the context of curriculum-specific humanities subjects. To the best of the authors' knowledge, no prior system integrates Braille OCR for examination scripts, domain-specific reference answer generation using fine-tuned language models, hybrid multi-metric evaluation, voice-oriented adaptive quizzes, and persistent learner modeling for Sri Lankan O-Level History. This gap directly motivates the integrated platform proposed in this paper.

III. METHODOLOGY

A. System Overview

The proposed platform is organized as a modular backend service composed of two primary subsystems that share a common data interface and deployment environment. Figure 1 illustrates the overall system architecture.

The first subsystem, the Braille Assessment Pipeline, accepts scanned or photographed Braille examination answer sheets as input. It performs image preprocessing, dot detection, spatial clustering, Unicode Braille decoding, and plain-text reconstruction. The resulting transcription is then parsed to separate examination questions from student answers. A fine-tuned language model subsequently generates reference answers, and a multi-metric scoring module produces a numerical score, grade classification, and natural-language explanatory feedback.

The second subsystem, the Adaptive Quiz Engine, is implemented as a FastAPI service that supports three primary assessment modes: generated multiple-choice quizzes, free-text answer evaluation, and adaptive quiz progression. It integrates a locally hosted large language model for dynamic question and distractor generation, a Sentence-BERT model for semantic answer evaluation, and a simplified item response theory model for real-time difficulty adaptation. A MongoDB-based persistence layer is used to manage learner sessions and performance history. The engine is designed to function as the reasoning layer behind a voice-first interface, where speech-to-text is used to capture learner responses and text-to-speech is used to deliver questions and feedback.

Both subsystems share the Sentence-BERT model for semantic similarity computation and are designed to be independently maintainable while remaining interoperable through their respective text-based interfaces.

B. Braille-to-Text Conversion Module

The Braille-to-text module accepts either a PDF examination paper or a single image as input. When a PDF is provided, it is first converted into high-resolution images at 200 DPI using PyMuPDF, producing one image per page. Each image is then processed independently through a five-step pipeline. In the preprocessing stage, images are converted to grayscale and resized if their width exceeds 2,500 pixels. Gaussian blur is applied to reduce sensor noise, and Otsu's thresholding is used to generate a binary image in which Braille dots appear as the foreground. Morphological opening is applied to remove residual speckle while preserving the structure of the dots.

Dot detection is performed using contour detection on the binary image. Contours with areas outside 20–400% of the median contour area are discarded, and the centroids of the remaining contours are recorded as dot coordinates. Geometric estimation is then used to determine characteristic dot spacing based on the median nearest-neighbor distance. The intra-cell column gap is computed from horizontal dot pairs.

Row clustering groups dot y-coordinates into row centers using a one-dimensional gap clustering algorithm. Each text row consists of three dot rows corresponding to the upper, middle, and lower positions of a Braille cell. Dot x-coordinates are subsequently clustered into columns and paired into left–right cell pairs using the intra-cell gap estimate. These cells are mapped to Unicode Braille characters, with word boundaries detected heuristically. The resulting sequence is decoded into English text using a lookup table, with capital and numeric markers handled in a stateful manner.

C. Model Fine-Tuning for Reference Answer Generation

A dataset of Sri Lankan O-Level History question–answer pairs was compiled from study materials and past examination papers. The Llama 3.1 8B Instruct model was selected due to its strong instruction-following capability and suitability for domain-specific fine-tuning. To reduce memory requirements, the model was loaded in four-bit quantized form using the NF4 scheme.

LoRA adapters were applied to both attention and feed-forward projection layers, with a rank of 16, a scaling factor of 32, and a dropout rate of 0.05. This configuration resulted in approximately 1.5% of the model parameters being trainable. Training was conducted using the Llama Instruct chat template with a maximum token length of 512. The dataset was split into 80% for training and 20% for validation.

The model was trained for three epochs with a per-device batch size of one and gradient accumulation to achieve an effective batch size of eight. A learning rate of 0.0002 with warmup and cosine scheduling was applied using the 8-bit AdamW optimizer. Gradient checkpointing was enabled to reduce memory usage, and validation loss was monitored to select the final model checkpoint.

D. Answer Evaluation Framework

During inference, the fine-tuned model generates a reference answer using greedy decoding with beam search of width four, zero temperature, and a repetition penalty of 1.2. Post-processing removes incomplete trailing sentences, producing a reference answer of up to approximately 200 words.

Student responses are compared against reference answers using three complementary metrics. Semantic similarity is computed as the cosine similarity between Sentence-BERT embeddings, capturing the overall meaning of the response. Keyword overlap is measured using a TF-IDF vectorizer, which calculates the proportion of high-weight reference keywords present in the student's answer, ensuring that domain-specific terminology is considered. Jaccard similarity evaluates surface-level consistency by computing the intersection over union of tokenized word sets after lowercasing and removing punctuation.

These metrics are combined into a final weighted score and scaled using a length penalty factor: 1.0 for answers exceeding 150 words, 0.75 for answers between 90 and 150 words, and 0.4 for shorter responses. The weighting scheme prioritizes semantic content while ensuring that keyword coverage and lexical alignment contribute to the final assessment.

E. Adaptive Voice Quiz Engine -Curriculum-Based Item Preparation

A structured item bank was constructed using chapter-based quiz data derived from the Sri Lankan O-Level History curriculum. Each record includes the chapter name, question, correct answer, supporting context, and a categorical difficulty label. Difficulty levels were standardized into three categories easy, medium, and hard and mapped to bounded

numerical values to support lightweight ability estimation during quiz progression.

F. Adaptive Session Management

When a learner initiates an adaptive quiz, the system initializes a session containing the username, selected chapter, current difficulty level, previously asked items, correctness counters, and an ability estimate initialized to a neutral value. Adaptive progression is governed by a simplified item response theory model. Each item is associated with a difficulty parameter b, while the learner state is represented by an ability estimate θ The probability of a correct response is given by:

$$P(\text{correct}) = \frac{1}{1 + e^{-a(\theta - b)}}$$

where a a represents item discrimination and is held constant for simplicity. After each response, the ability estimate is updated using a bounded step rule: correct responses increment θ, while incorrect responses decrement it within a fixed interval to prevent instability. Difficulty is increased when the learner correctly answers at least two out of three consecutive questions at a given level, enabling gradual adaptation suitable for audio-based interaction.

G. AI-Assisted Question and Distractor Generation

The adaptive quiz primarily delivers multiple-choice questions from the item bank. When distractor options are not predefined, a locally hosted large language model generates three plausible, but incorrect alternatives related to the same historical topic and period. A lock-based access strategy is employed to prevent concurrent inference requests from corrupting the model state. If generation fails, safe placeholder options are used to ensure session continuity, reflecting the design principle that accessibility-oriented systems should prioritize uninterrupted interaction.

H. Answer Normalization and Semantic Evaluation

For free-text responses, the backend performs normalization of historical answer variants prior to evaluation. For example, ruler names expressed in Roman numeral, ordinal, or cardinal forms such as Mihindu V, Mihindu 5, or Mihindu fifth are standardized before applying Sentence-BERT-based similarity scoring. This normalization is particularly important in voice-based contexts, where speech recognition may produce semantically equivalent but textually different outputs.

F. Persistence and Learner State Management

All adaptive quiz interactions are stored in MongoDB, enabling persistence of active sessions, previously asked items, current difficulty levels, and evolving ability estimates. Upon completion of a session, a summary record is generated, including the final ability estimate, total number of questions attempted, final difficulty level, and completion timestamp. These records are aggregated within the user profile module to support longitudinal performance tracking, teacher-facing progress review, and future personalization.

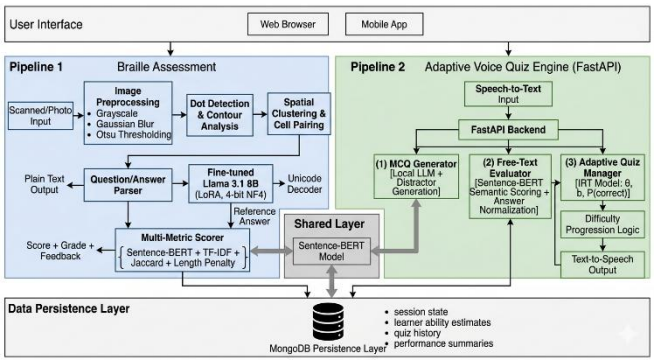

Fig. 1. Architecture of the system

IV. RESULTS AND ANALYSIS

A. Braille-to-Text Conversion Performance

The Braille-to-text module was tested on scanned and photographed answer sheets. The system successfully detected dot patterns, segmented text rows, identified cell boundaries, and decoded Unicode Braille characters to English for all test pages. Morphological preprocessing and gap-based clustering handled common imaging artifacts such as page tilt, uneven illumination, and minor dot coalescence.

Qualitative analysis confirmed correct handling of capital letter markers, number prefixes, and inter-word spaces in the majority of cases. Errors primarily occurred on pages with very faint dots or in dense paragraphs where adjacent rows interfered during clustering. The 200 DPI PDF-to-image conversion provided sufficient resolution for reliable dot detection.

Table I summarizes key performance observations for the Braille-to-text module.

TABLE I PERFORMANCE OBSERVATIONS OF BRAILLEN TO TEXT MODULE

Metric	Observed Behavior
Dot Detection Accuracy	High for standard ink or pressure errors with faint dots
Row Segmentation	Accurate in > 95% of lines
Cell Pairing	Robust; minor errors in dense handwriting
Capital/Number Handling	Correct in majority of cases
Word Boundary Detection	Correct for most paragraphs

B. Reference Answer Generation

The fine-tuned Llama 3.1 model generated coherent, factually grounded reference answers for O-Level History questions. Generated answers consistently included key events, figures, and dates within the 200-word limit. Training and validation loss curves indicated proper generalization without overfitting.

LoRA fine-tuning successfully injected curriculum-specific knowledge while preserving base model fluency and instruction-following ability. Post-processing removed trailing incomplete sentences, improving readability.

Table II shows qualitative examples of reference answer generation.

TABLE II QUALITATIVE EXAMPLES OF REFERENCE ANSWER GENERATION

Question	Key Historical Concepts	Generated Answer Quality
Colonial history of Sri Lanka	Portuguese, Dutch, British	High:correct events and dates, coherent
Anuradhapura period kings	Devanampiya Tissa, Dutugemunu	High:factual and concise

C. Answer Evaluation Performance

The three-metric evaluation framework, which combines semantic similarity, keyword overlap, and Jaccard similarity, produced interpretable scores that reflected the degree of conceptual alignment between student answers and reference answers. High semantic similarity scores were observed when students accurately explained historical facts, while moderately high scores were assigned to responses that were factually relevant but expressed in different phrasing. The keyword overlap metric captured the presence of domain-specific terms, including ruler names, historical events, and dates, enhancing the granularity of the evaluation. Jaccard similarity provided a surface-level consistency check between tokenized word sets, complementing the deeper semantic assessment. To ensure that trivially brief responses did not receive disproportionately high scores, a length penalty was applied, effectively scaling the final score in proportion to the completeness and depth of the answer.

Table III shows sample evaluation outcomes.

TABLE III EVALUATION OUTCOMES

Student Answer Type	Semantic (%)	Keyword (%)	Jaccard (%)	Final Weighted Score
Complete& accurate	92	88	85	90
Partially correct	70	60	65	66
Minimal content	40	30	35	34

The natural-language feedback accurately reflected scoring outcomes, highlighting strong coverage for high-scoring answers and identifying missing details for mid-range answers, consistent with O-Level marking standards.

D. Adaptive Quiz Component

The adaptive quiz component was evaluated using the structured item bank derived from the Sri Lankan O/L History curriculum, including chapter-based questions and past-paper content. The pipeline incorporated curriculum-constrained question delivery, rule-based adaptive progression, AI-assisted distractor generation, and voice-oriented interaction through integration with speech recognition and synthesis services. The progression mechanism promoted difficulty reliably after two correct answers in three attempts, and the lock-based inference strategy prevented concurrent generation failures during multi-user testing.

Effective evaluation of the adaptive quiz component requires pedagogical expert assessment rather than a single computational metric, because its educational value depends on syllabus alignment, appropriateness of question difficulty, usefulness of feedback, and suitability for audio-first interaction. Structured evaluation with Sri Lankan O/L History teachers and usability studies with visually impaired students remain the appropriate method for validating pedagogical effectiveness, and this is identified as a primary direction for future work.

TABLE IV EVALUATION METHODS OF ADAPTIVE QUIZ

Evaluation Aspect	Evaluation Method
Curriculum relevance	Teacher review
Difficulty appropriateness	Expert evaluation
Accessibility support	Usability observation

E. Overall Analysis

The integrated platform demonstrated robust performance across all processing stages. The Braille-to-text module consistently produced accurate English transcriptions suitable for automated evaluation. The reference answer generation component effectively aligned with curriculum expectations, while the weighted scoring framework captured semantic, lexical, and domain-specific similarities with precision. The adaptive quiz engine ensured smooth difficulty progression and uninterrupted session management. Additionally, the modular system design facilitated targeted debugging and supports future scalability, making it well-suited for integration into accessible educational platforms.

V. CONCLUSION

This paper presents an integrated AI-powered platform designed to address two key accessibility barriers faced by visually impaired O-Level History students in Sri Lanka: the lack of efficient automated evaluation for Braille examination scripts, and the absence of curriculum-specific interactive self-study tools.

The Braille Assessment Pipeline converts scanned answer sheets into English transcriptions using image processing, eliminating the need for specialized hardware. These transcriptions are evaluated using a fine-tuned Llama 3.1 model and a weighted combination of semantic, keyword, and lexical similarity metrics. The Adaptive Quiz Engine offers voice-compatible question generation, semantic free-text evaluation, real-time difficulty adaptation, and persistent learner modeling, all implemented through a modular FastAPI service.

By combining these components, the platform enables faster, more consistent examination evaluation and provides an independent, adaptive study tool optimized for audio-first interaction. Parameter-efficient fine-tuning via LoRA allows domain-specific adaptation without large-scale computational resources, making the approach practical and reproducible in student-led research contexts.

Key limitations include the lack of contracted Braille recognition, evaluation scoring thresholds determined by expert judgment rather than empirical calibration with labeled student data, and a rule-based adaptive progression model that does not yet incorporate full psychometric calibration. These limitations highlight clear directions for future work.

VI. FUTURE WORK

Future development will proceed along five parallel directions. First, the Braille recognition module will be extended to support contracted Braille, which is widely used in examination contexts and requires look-ahead, context-sensitive decoding beyond the current cell-by-cell approach. Second, the fine-tuning dataset will be expanded to cover all chapters of the Sri Lankan O/L History curriculum, improving reference answer coverage and factual accuracy. Third, a structured evaluation study involving qualified History teachers and visually impaired students will enable empirical calibration of scoring thresholds, metric weights, and adaptive progression parameters.

Fourth, the item bank will be expanded and better balanced across all O/L History chapters, while the rule-based learner modeling will be enhanced using advanced adaptive assessment techniques to improve personalization and reliability. Fifth, the platform will be integrated with a voice-first front-end incorporating speech-to-text and text-to-speech services, improved recognition of local historical terminology, and a multilingual extension framework — enabling end-to-end testing from spoken question presentation to Braille response capture, automated scoring, and audio feedback delivery in realistic educational settings.

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